

Genetic and biochemical interactions of yeast DNA helicases

David G. Nickens, Matthew L. Bochman^{*}

Molecular & Cellular Biochemistry Department, Indiana University, Bloomington, IN 47405 USA

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ABSTRACT

DNA helicases function in many types of nucleic acid transactions, and as such, they are vital for genome integrity. Although they are often considered individually, work from many groups demonstrates that these enzymes often genetically and biochemically interact *in vivo*. Here, we highlight methods to interrogate such interactions among the PIF1 (Pif1 and Rrm3) and RecQ (Hrq1 and Sgs1) family helicases in *Saccharomyces cerevisiae*. The interactions among these enzymes were investigated *in vivo* using deletion and inactivation alleles with a gross-chromosomal rearrangement (GCR) assay. Further, wild-type and inactive recombinant proteins were used to determine the effects of the helicases on telomerase activity *in vitro*. We found that synergistic increases in GCR rates often occur in double vs. single mutants, suggesting that the helicases function in distinct genome integrity pathways. Further, the recombinant helicases can function together *in vitro* to modulate telomerase activity. Overall, the data suggest that the interactions among the members of these DNA helicase families are multipartite and argue for a comprehensive systems biology approach to fully elucidate the physiological interplay between these enzymes.

1. Introduction

DNA helicases are enzymes that use the combination of DNA binding and ATPase activities to drive translocation along DNA and DNA unwinding [1]. Though conceptually simple, this process forms the basis of many essential DNA transactions in the cell, from replication to repair to transcription. As such, DNA helicases are found in all species and known to be key players in genome maintenance. Further, they are evolutionarily conserved and can be divided into at least seven distinct helicase superfamilies [2].

Two such conserved families of DNA helicases are the PIF1 and RecQ helicases. PIF1 family enzymes are found in bacteria and eukaryotes, though have almost exclusively been studied in the latter, especially the budding yeast *Saccharomyces cerevisiae* [3]. To date, the *S. cerevisiae* Pif1 helicase has been shown to participate in mitochondrial DNA maintenance [4,5], telomere maintenance [6–9], multiple processes during DNA replication [10–14], repairing single-stranded nicks by break-induced replication [15], and resolving G-quadruplex (G4) structures [16]. *S. cerevisiae* also encodes a second PIF1 family helicase named Rrm3 that functions in many of the same processes as Pif1, though they often display opposing activities [3,17].

Likewise, the budding yeast genome encodes two RecQ family helicases, Hrq1 and Sgs1, which are homologs of the disease-linked human RECQL4 and BLM enzymes, respectively [1,9,18–23]. Overall, RecQ helicases are also known to be multifunctional genome caretakers, participating in DNA replication, recombination, and repair, among other processes [24–26]. With respect to Hrq1 and Sgs1, these enzymes specifically have roles in DNA inter-strand crosslink repair [9,20,27], telomere maintenance [9,28–30], and DNA repair by recombination [21,31–34], as well as resolve non-canonical DNA secondary structures such as G4s [20,35–38].

In addition to functioning in similar pathways that underpin genome integrity, the PIF1 and RecQ helicases are known to genetically and physically interact. An early observation of this was made by the Rothstein lab when they reported that Pif1 genetically interacts with Sgs1 and Top3 (topoisomerase III) downstream of homologous recombination [39]. Pif1-RecQ interactions have also been reported by the Kolodner and Zakian labs [40,41]. More recently, our group has demonstrated that Hrq1 genetically and functionally interacts with Pif1 during telomere maintenance [9,28,29] and that both RecQ helicases, Hrq1 and Sgs1, genetically interact [18,42]. Here, we provide new data to further support these interactions, focusing on the methods used to

Abbreviations: G4, G-quadruplex; 5-FOA, 5-fluoroorotic acid; GCR, gross-chromosomal rearrangement; Chr V-L, left arm of chromosome V; 5-FOA+Can, medium containing 5-FOA and canavanine; IQR, inter-quartile range; DSB, double-strand break.

^{*} Corresponding author.

E-mail address: bochman@iu.edu (M.L. Bochman).

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gather the data, which are amenable to investigating any such *S. cerevisiae* helicases.

2. Materials and methods

2.1. Yeast strains, media, and other reagents

All strains (Table 1) were created by standard methods and are derivatives of the YPH500 background (*MAT α ura3-52 lys2-801_amber ade2-101_ochre trp1 Δ 63 his3 Δ 200 leu2 Δ 1*) [43]. Details concerning their construction can be found in [9,16,42,44] and are available upon request. Cells were grown in standard *S. cerevisiae* media (e.g., YPD or synthetic drop out media) at 30 °C with aeration unless otherwise noted. Synthetic drop-out mix lacking arginine, uracil, and yeast nitrogen base (SC-Arg-Ura) was from US Biologicals. 5-fluoroorotic acid (5-FOA) was purchased from Toronto Research Chemicals, canavanine sulfate was from Sigma, and all other chemical reagents were of the highest grade available. All restriction enzymes were from New England Biolabs, and all oligonucleotides were purchased from Integrated DNA Technologies.

2.2. Gross-chromosomal rearrangement (GCR) assay

The *S. cerevisiae* GCR assay has been the subject of many excellent reviews (e.g., [45–47]), including those that provide detailed protocols. To avoid redundancy, we here provide practical methods and tips for successfully executing the assay and interpreting the results. Further, several different versions of the GCR assay have been developed [48], but we will focus on the original assay [49] and its requirements below.

The classical GCR assay monitors the function of the *CAN1* and *URA3* genes on the left arm of Chromosome V (Chr V-L) (Fig. 1). This portion of the genome natively contains the *CAN1* locus, but *URA3* must be placed in proximity to *CAN1* in an otherwise Ura⁻ strain. This generally occurs by insertion of the *URA3* gene or a *URA3*-containing selectable marker cassette (e.g., [50]) into the nonessential *HXT13* locus, which is approximately 8.5 kb away from *CAN1* toward the telomere. This creates what we refer to as our wild-type GCR strain MBY49 (*MAT α ura3-52 lys2-801_amber ade2-101_ochre trp1 Δ 63 his3 Δ 200 leu2 Δ 1 hxt13::URA3*), which is the parental strain of the helicase mutants shown in Table 1.

2.2.1. Pouring plates containing 5-FOA and canavanine (5-FOA + Can)

The GCR assay takes advantage of counter-selection conferred by the

URA3 and *CAN1* genes using media containing 5-FOA and canavanine, respectively. These 5-FOA + Can plates can be poured ahead of time and stored in the dark in sealed plastic bags at 4 °C for at least a month without losing their effectiveness. The following is a protocol to make 1 L of 5-FOA + Can agar to pour plates.

1. Add 24 g agar and 20 g glucose to 750 mL Milli-Q H₂O in a 2-L flask containing a magnetic stir bar. Stir until the sugar dissolves, and then autoclave to sterilize.
2. In a separate flask or beaker containing a magnetic stir bar, add 1 g 5-FOA, 60 mg canavanine sulfate, 2 g drop-out mix synthetic minus arginine, uracil without yeast nitrogen base (powder; US Biological), 50 mg uracil, 6.7 g yeast nitrogen base without amino acids, and 250 mL Milli-Q H₂O.
3. Stir with gentle heating until all of the reagents go into solution.
4. Filter sterilize through a 0.2- μ m membrane, and keep the solution warm in a 37 or 42 °C water bath.
5. Cool the autoclaved agar solution to ~ 65 °C while gently stirring. Stop the stirring, and slowly add the pre-warmed minus arginine, uracil solution. Stir slowly to evenly mix everything, and pour into 100-mm plastic culture dishes (~25 mL per plate). If bubbles form during the mixing or pouring processes, mist 95% ethanol over the flask or plate(s) using a spray bottle to pop the bubbles.

2.2.2. Performing the GCR assay

The following is based on the protocols published by the Kolodner lab [45–47]. We have found that the best advice when performing this assay with many individual cultures is to vortex the cultures tubes each and every time before removing any cells for the steps described below. Otherwise, the cells in your saturated cultures will fall out of suspension, and your counts will not be accurate.

1. Streak your strain(s) of interest for single colonies on YPD plates and incubate at 30 °C.
2. Set up 21 (three sets of seven for each strain) 5-mL YPD overnight cultures by inoculating the tubes with one colony each, and incubate at 30 °C with aeration for ~ 48 h to saturate growth. In general, we inoculate seven independent biological replicates each day for 3 days and perform three back-to-back-to-back assays, as described below, on these sets of cultures once saturated.
3. Make serial dilutions for each culture in sterile 96-well plates:
 - a. Add 90 μ L sterile H₂O to six wells for each culture.
 - b. Using a P20 pipette, add 10 μ L culture to well 1, mix by vigorously pipetting up and down, add 10 μ L to well 2. We do not recommend the use of a multi-channel pipette for these dilutions because, in our experience, they are less accurate than single-channel pipettes for this type of work. Further, pipette size (P10 vs. P20 and P100 vs. P200), the choice of pipette tip brands, and pipette calibration can impact the accuracy of this step and the ultimate viable cell count. Thus, larger volumes (100 μ L culture and 900 μ L H₂O) and using a vortex to thoroughly mix the dilutions can increase accuracy.
 - c. Continue this serial dilution through well 6 to make the 10⁻⁶ dilution.
 - d. Plate all 100 μ L from well 6 onto a YPD plate, yielding a final dilution factor of 10⁻⁷, and incubate the plates for 4 days at room temperature.
 - e. Count and record the number of colonies on each plate
4. Pellet 2 mL of cells for each culture, and discard the supernatant.
5. Resuspend the cells in 200 μ L sterile H₂O, plate them onto 5-FOA + Can plates, and incubate for 4–5 days at 30 °C. It helps to incubate the plates in sealed plastic bags to avoid the media drying out.
6. Count and record the number of colonies on each 5-FOA + Can plate. This value = r.
7. Calculate N (should be in the 10¹⁰ range) using the following table:

Table 1

Saccharomyces cerevisiae strains used in this study.

Strain name	Genotype	Source
KP311	<i>pif1-m2 rrm3::His3MX6</i>	[16]
KP313	<i>pif1-m2</i>	[16]
KP351	<i>pif1-m2 sgs1::his5+</i>	[16]
MBY49	Wild-type ^a	[16]
MBY79	<i>rrm3::His3MX6</i>	[16]
MBY132	<i>sgs1::his5+</i>	[16]
MBY161	<i>hrq1::His3MX6</i>	[9]
MBY164	<i>hrq1::His3MX6 pif1-m2</i>	This study
MBY167	<i>hrq1::TRP1 rrm3::His3MX6</i>	This study
MBY168	<i>hrq1::TRP1 sgs1::his5+</i>	This study
MBY280	<i>hrq1::hrq1-K318A-NatMX</i>	[9]
MBY347	<i>hrq1::hrq1-K318A-NatMX sgs1::his5+</i>	This study
MBY382	<i>hrq1::hrq1-K318A-NatMX pif1-m2</i>	This study
MBY384	<i>hrq1::TRP1 pif1-m2 sgs1::His3MX6</i>	This study
MBY385	<i>hrq1::TRP1 pif1-m2 rrm3::His3MX6</i>	
MBY386	<i>hrq1::hrq1-K318A-NatMX sgs1::sgs1-K706A-TRP1</i>	This study
MBY387	<i>sgs1::sgs1-K706A-TRP1</i>	[9]
MBY446	<i>hrq1::His3MX6 sgs1::sgs1-K706A-TRP1</i>	This study
MBY452	<i>pif1-m2 sgs1::sgs1-K706A-TRP1</i>	This study
MBY457	<i>hrq1::hrq1-K318A-NatMX pif1-m2 sgs1::sgs1-K706A-TRP1</i>	This study

^a The wild-type strain is an *hxt13::URA3* derivative of YPH500 (*MAT α ura3-52 lys2-801_amber ade2-101_ochre trp1 Δ 63 his3 Δ 200 leu2 Δ 1*) [43].

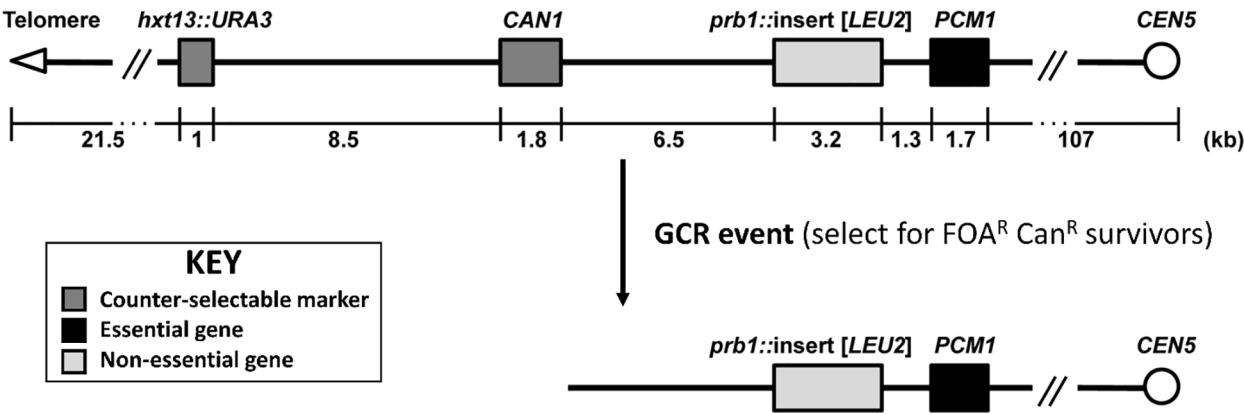


Fig. 1. Schematic of the GCR assay. A depiction of Chr V-L of a strain built for the GCR assay is shown, including the counter-selectable markers, a nearby non-essential locus (*PRB1*) where various DNA inserts can be added to gauge their effects on the GCR rate (e.g., G4 motifs [16]), and the essential gene closest to the telomere (*PCM1*) on this arm of the chromosome. After a GCR event, the counter-selectable markers are lost (as shown) and/or mutated such that they no longer encode functional proteins.

Plate	Colonies on YPD x	Vol plated on 5-FOA+Can x	Dilution=	N
1		2 mL	10 ⁷	
2		2 mL	10 ⁷	
3		2 mL	10 ⁷	
4		2 mL	10 ⁷	
5		2 mL	10 ⁷	
6		2 mL	10 ⁷	
7		2 mL	10 ⁷	

8. Copy and paste r and N into the proper fields at the Fluctuation Analysis Calculator (FALCOR) webserver (<https://lianglab.brocku.ca/FALCOR/>) [51]. Group the data by 7, change the rate to 10⁻⁹ mutations per generation, use the MSS Maximum Likelihood Method for the most accurate results, and hit Submit to calculate your mutation rate (M) and 95% confidence interval.
9. Most publications (e.g., [52,53]) report the rate as the median rate and include the 95% confidence interval for comparison between strains. If calculating the mean GCR rate of a strain as we do here, average the mean rates across your ≥ 3 sets of experiments. If analyzing the data in this way, one should consider each data set in relation to the others before combining them. The general criterion for combining data from individual experiments using a single strain is that individual measurements should be within the biological variability (~2-fold) for a GCR assay. A discussion of outliers in the data and how to identify them is included below in Section 2.2.4. When comparing the differences between GCR rates, we use Student's *t*-test in GraphPad Prism 8, not assuming a consistent standard deviation and with $\alpha = 0.05$, to calculate *p* values.

2.2.3. What happens if the GCR rate is very low?

In strains with low GCR rates, such as wild-type cells, plating 2 mL of cells per culture will often yield zero colonies on a plate. In such cases, larger cultures (up to 50 mL) are recommended, and more cells must be plated on 5-FOA + Can plates. However, we still recommend that only 2 mL of culture be added per plate and simply that multiple plates per culture be used. This is because plating the cells from 2 mL of saturated liquid culture leaves a visible film of non-growing cells on the plate, in addition to the 5-FOA^R Can^R colonies that are sought. When additional cells are plated, the thickness of this layer of cells can become a problem for the 5-FOA + Can counter-selection and also for how long it takes the plates to dry after cell addition.

2.2.4. Can the GCR results be trusted?

GCR events happen at an extremely low level, even in most mutant

strains [16,45–47], so the number of colonies being counted on plates is consequently also low. Thus, despite multiple cultures being examined per experiment (three sets of seven as described above), the GCR rates can be heavily skewed by a single plate of outliers. These outliers are often plates containing many colonies due to jackpot mutations, *i.e.*, mutants that arise by chance early during the growth of a culture and thus represent a larger number of cells in the final population than mutations that occur later during growth of the cell population [54,55]. Indeed, in strains with low GCR rates that are only slightly greater than wild-type (e.g., *hrq1*Δ; Table 2), jackpotting is usually obvious. In a single experiment with seven cultures, most will yield single-digit numbers of 5-FOA^R Can^R colonies, but an outlier plate will display hundreds of colonies. That said, in our experience, outliers are rare. If

Table 2
Normalized GCR rates.

Strain	Mean GCR rate ± standard deviation ^d
Wild-type ^a	1.0 ± 0.5 ^b
Single mutants	
<i>hrq1</i> Δ	4.3 ± 2.0 ^c
<i>hrq1-K318A</i>	5.0 ± 3.0 ^c
<i>pif1-m2</i>	76 ± 8.0 ^b
<i>rrm3</i> Δ	6.0 ± 5.0 ^b
<i>sgs1</i> Δ	16 ± 5.0 ^b
<i>sgs1-K706A</i>	14 ± 3.2 ^c
Double mutants	
<i>hrq1</i> Δ <i>pif1-m2</i>	12 ± 6.8
<i>hrq1</i> Δ <i>rrm3</i> Δ	32 ± 7.3
<i>hrq1</i> Δ <i>sgs1</i> Δ	6.0 ± 5.0
<i>hrq1</i> Δ <i>sgs1-K706A</i>	6.7 ± 0.9
<i>hrq1-K318A</i> <i>pif1-m2</i>	53 ± 19
<i>hrq1-K318A</i> <i>sgs1</i> Δ	34 ± 14
<i>hrq1-K318A</i> <i>sgs1-K706A</i>	7.7 ± 5.9
<i>pif1-m2</i> <i>rrm3</i> Δ	210 ± 32 ^b
<i>pif1-m2</i> <i>sgs1</i> Δ	220 ± 75 ^b
<i>pif1-m2</i> <i>sgs1-K706A</i>	340 ± 140
Triple mutants	
<i>hrq1</i> Δ <i>rrm3</i> Δ <i>pif1-m2</i>	490 ± 13
<i>hrq1</i> Δ <i>pif1-m2</i> <i>sgs1</i> Δ	3.4 ± 1.6
<i>hrq1-K318A</i> <i>pif1-m2</i> <i>sgs1-K706A</i>	160 ± 68

^a Without normalization, the wild-type GCR rate was 1.5 ± 0.7 × 10⁻¹⁰ events/generation. Outliers were omitted from these calculations as described in Section 2.2.4.

^{b,c} Data are from [16] (b) or [9] (c).

^d The mean rates were calculated by determining the GCR rates for each of ≥ 3 sets of experiments and averaging those values.

such events occur in every data set for a given strain, for instance, this could be indicative of batch-to-batch variability in media preparation if 5-FOA + Can plates are made by multiple individuals, or the strain may have an underlying genomic instability unrelated to the GCR assay.

Data points that are outliers can be identified using software such as GraphPad Prism using box and whisker plots, with points less than the value of the 25th percentile (lower quartile) minus 1.5 times the interquartile range (IQR) or greater than the value of the 75th percentile (upper quartile) plus 1.5IQR being outliers [56]. If GraphPad Prism or similar software is not available, web tools (e.g., <https://miniwebtool.com/outlier-calculator/>) are also freely available to identify outliers using the same method. To remove bias, every GCR experiment is analyzed for outliers, not just ones that “seem” to have plates with too many or too few colonies based on previous experience with a strain or previously reported GCR rates. When outliers are identified, they are omitted from the overall data sets when calculating GCR rates (see above). However, if reporting data as described in Step 9 of Section 2.2.2 above, outliers should not be eliminated from the data when determining the median and reporting 95% confidence intervals.

2.3. In vitro telomerase assays

2.3.1. Proteins

Recombinant *S. cerevisiae* Hrq1, Pif1, and Sgs1 were generated as previously described [9,19,27,28], and these previously published preparations were used here. Briefly, Hrq1 and Sgs1 were over-expressed in baculovirus-infected insect cells and purified by nickel affinity (Hrq1) or anti-FLAG affinity (Sgs1) chromatography. Pif1 was over-expressed in *Escherichia coli* cells and purified by cobalt affinity chromatography. Purification samples were analyzed by SDS-PAGE and Coomassie staining to identify peak fractions, which were pooled and dialyzed against storage buffer. Protein concentrations were quantified on SYPRO orange (Sigma)-stained SDS-PAGE gels using BSA dilutions to plot a standard curve. Gel imaging and analysis were performed using a Typhoon 9210 Variable Mode Imager (Amersham Biosciences). Catalytically inactive Hrq1-K318A and Pif1-K264A mutants were over-expressed and purified as described for their wild-type counterparts.

Telomerase-enriched cell extracts were prepared by DEAE affinity chromatography using clarified lysates [7,28]. Briefly, the telomerase holoenzyme was over-expressed by galactose induction, and cells were harvested and frozen in liquid nitrogen. After cyro-lysis, the thawed lysate was clarified by centrifugation, incubated with DEAE-Sepharose Fast Flow (GE Healthcare) and washed with a low-salt buffer. Telomerase was then eluted with a high-salt buffer, concentrated by ultrafiltration, and flash frozen.

2.3.2. Telomerase primer extension assay

The method used to investigate the effects of Pif1 on telomerase is not new [7,28], and a detailed protocol to measure the telomerase-modulating effects of Hrq1 and Pif1 together was recently published [57]. In brief, an oligonucleotide comprised of the telomeric repeat sequence is mixed with a telomerase-enriched extract, dGTP, and [α -³²P]TTP – with or without the addition of one or more recombinant helicases – and primer extension products are visualized by phosphor-imaging after electrophoresis on large denaturing gels. To avoid simply repeating the protocol here, we instead highlight important steps, controls, and tips for troubleshooting.

1. Each telomerase preparation should be titrated to standardize activity levels before use in experiments. We always reserve 10% of each preparation for these comparisons between old and new batches.
2. Control experiments should be performed to ensure that extension products are generated by telomerase and not another polymerase activity. RNase A addition to the reactions should inhibit telomerase activity by degrading the *TLC1* RNA template subunit of the

telomerase holoenzyme. Similarly, because the *S. cerevisiae* telomere repeat sequence is TG₁₋₃, only dGTP and dTTP are needed to produce extension products. Control reactions lacking one or both of these added dNTPs should show short or no extension products.

3. High-percentage polyacrylamide gels often crack while drying because too much TEMED is used to polymerize the gels. When pouring the 16% 19:1 acrylamide/bisacrylamide gels containing 6 M urea used here, a minimal amount of TEMED should be used (i.e., 10 μ L TEMED for every 100 mL of degassed acrylamide solution). This yields gels that are flexible and resistant to cracking during vacuum drying or freezing, with no methanol or glycerol pretreatment necessary.

3. Results and discussion

3.1. PIF1 and RecQ helicases genetically interact to impact genome stability

Early uses of the GCR assay in yeast revealed that mutations in *PIF1* or *SGS1* increase GCR rates [58,59], consistent with their multiple known roles in genome maintenance [1]. Similarly, mutations in *RRM3* and *HRQ1* have also been shown to increase GCR rates, and a double *rrm3* Δ *sgs1* Δ mutant displays a large synergistic increase in GCR rate compared to the single mutant [52]. To follow up on this work, we performed the classical GCR assay [49] to compare the effects of single and multiple PIF1 and RecQ helicase mutants on genome integrity (Table 2).

In line with reported findings [52,58,59], all single mutant strains displayed elevated GCR levels relative to wild-type cells, though the rates in the *hrq1* mutants and the *rrm3* Δ strain were only marginally increased (Table 2; $p < 0.05$). As shown previously [52], the GCR rate in many double mutants was synergistically increased related to the single mutants, such as *hrq1* Δ *rrm3* Δ , *pif1*-*m2* *rrm3* Δ , and *pif1*-*m2* *sgs1* Δ (). However, this pattern did not hold true for all double mutants. The GCR rate in *hrq1* Δ *pif1*-*m2* cells was ~6-fold lower than in the *pif1*-*m2* strain, and the rate was also > 2-fold decreased in *hrq1* Δ *sgs1* Δ cells compared to the *sgs1* Δ mutant. A similar phenomenon was noted with triple mutants: the GCR rate in the *hrq1* Δ *rrm3* Δ *pif1*-*m2* strain was > 2-fold above that of any of the double mutants containing combinations of *hrq1* Δ , *rrm3* Δ , and *pif1*-*m2*, but the GCR rate in the *hrq1* Δ *pif1*-*m2* *sgs1* Δ triple mutant was nearly back to baseline.

The reasons for these differences are not clear. One hypothesis for the large elevation in the GCR rate in *pif1* strains is that the absence of nuclear Pif1 activity leads to defects in Okazaki fragment maturation, a process that occurs numerous times every cell cycle across the relevant portion of Chr V-L monitored by the GCR assay (Fig. 1). The large number of Okazaki fragments processed over multiple cell generations during the course of the assay provides ample opportunity for GCR events to occur in *pif1*-*m2* cells. Alternatively, however, in the absence of nuclear Pif1, perhaps Hrq1 helicase activity creates a toxic DNA structure that leads to the elevated GCR rate in *pif1*-*m2* mutant (Table 2). This would explain the decrease in the GCR rate of the *hrq1* Δ *pif1*-*m2* cells, as deletion of *HRQ1* would prevent the toxic structure from forming. Although Hrq1 is known to function in DNA inter-strand crosslink repair and telomere maintenance [9,27,28], it is also associated with numerous other processes *in vivo* [42,44], and a mechanistic understanding of the roles of Hrq1 is still largely lacking. Thus, the identity of this potential toxic DNA structure and the physiological process during which it may form remain to be investigated.

Why does *HRQ1* deletion also decrease the GCR rate of the *pif1*-*m2* *sgs1* Δ strain? One could invoke the same toxic DNA structure hypothesis as above to explain this phenomenon, but the reason is likely to be different. The genetic interactions between *PIF1* and *SGS1* [39], as well as *HRQ1* and *SGS1* [9,18,27,42], are known to be complex, and each of these enzymes plays multiple roles *in vivo* to support genome integrity. Thus, the tripartite genetic interactions between these helicases are

expected to be even more complex and potentially vary based on variables such as the phase of the cell cycle and presence or type of DNA damage. Further work is needed to determine the binding sites of Hrq1, Pif1, and Sgs1 genome wide, their physical interactomes, and their genetic interaction landscapes in wild-type, single-, and double-helicase mutant cells. Such a systems biology approach may be the only way to truly tease apart these intricacies.

3.1.1. The effects of catalytically inactive helicase mutants on GCR rates

It is known that catalytically inactive alleles of DNA helicases are often dominant negative [60], and we have found this to be the case with *hrq1* Δ vs. *hrq1-K318A* [9,27]. Thus, we also wanted to test the effects of the *hrq1-K318A* allele on the GCR rates of PIF1 and RecQ mutants. As previously shown [9], the GCR rate of the *hrq1-K318A* mutant is similar to that of the *hrq1* Δ strain (Table 2). However, unlike above with *hrq1* Δ , when this mutation is combined with *pif1-m2*, the GCR rate does not significantly differ from that of *pif1-m2* alone ($p = 0.12$). The reason for this dichotomy is unclear, but it is known that *hrq1* Δ and *hrq1-K318A* display differences in how DNA double-strand breaks (DSBs) are repaired after GCR events [9]. Most DSBs are healed by telomere addition in *hrq1* Δ cells, but very few telomere additions occur in the *hrq1-K318A* mutant. This is likely because Pif1 is present in the *hrq1* Δ background and can stimulate telomerase activity, but Hrq1-K318A and Pif1 functionally interact as a strong telomerase inhibitor [28].

Regardless, we additionally sought to analyze the effects of the inactive *sgs1-K706A* mutation. Again, as shown previously, the GCR rate of the *sgs1-K706A* strain was similar to that of *sgs1* Δ [9], and the double mutant with *pif1-m2* displayed a synergistic increase in GCR rate, phenocopying the *pif1-m2 sgs1* Δ double mutant. Thus, the genetic interactions between the deletion and inactive alleles of the two yeast RecQ helicases with *pif1-m2* are quite distinct, with the *hrq1-K318A* mutation curiously not behaving like *hrq1* Δ as noted above.

We previously demonstrated that deletion and catalytically inactive alleles of the yeast RecQ helicases differentially genetically interact using synthetic genetic array analyses [42], so we wanted to further measure the effects of the *hrq1* Δ *sgs1* Δ , *hrq1* Δ *sgs1-K706A*, *hrq1-K318A sgs1* Δ , and *hrq1-K318A sgs1-K706A* double mutants on the GCR rate here. In three out of the four mutants, the GCR rate was not significantly different from the *hrq1* Δ or *hrq1-K318A* single mutants (Table 2; $p \geq 0.13$). The one exception to this was the *hrq1-K318A sgs1* Δ strain, with a GCR rate 34-fold over wild-type cells. These results were somewhat unexpected because the *hrq1* Δ *sgs1* Δ , *hrq1-K318A sgs1* Δ , and *hrq1-K318A sgs1-K706A* strains display similar levels of DNA damage sensitivity, but the *hrq1* Δ *sgs1-K706A* mutant is less sensitive than the others and actually displays resistance to hydroxyurea compared to the wild-type [9]. Here though, the *hrq1-K318A sgs1* Δ strain differed from the other mutants, not the *hrq1* Δ *sgs1-K706A* strain.

Finally, the GCR rate of the *hrq1-K318A pif1-m2 sgs1-K706A* triple mutant was also measured. It was significantly greater than that of the *hrq1* Δ *pif1-m2 sgs1* Δ mutant ($p = 0.016$) but not from either the *hrq1-K318A pif1-m2* ($p = 0.058$) or *pif1-m2 sgs1-K706A* ($p = 0.12$) double mutant strains (Table 2). Overall, these data once again indicate complex genetic interactions between *HRQ1*, *PIF1*, and *SGS1*. Considering that the *hrq1-K318A* and *sgs1-K706A* alleles yield wild-type levels of inactive proteins, the data presented here may also argue for complex physical interactions, either via direct helicase-helicase contacts or a progression of helicases working on the same substrate, one generating a DNA structure that requires the next to resolve.

3.2. PIF1 and RecQ helicases interact to regulate telomerase activity

One class of loci where the PIF1 and RecQ helicases are all known to act is telomeres. In *S. cerevisiae*, the PIF1 family helicases Rrm3 and Pif1 promote telomeric DNA replication [61] and the regulation of telomerase [6,7,28], respectively, and Hrq1 and Sgs1 are also known to

function in multiple aspects of telomere maintenance [9,28–30,62–65]. Thus, these enzymes may functionally interact when localized to the same telomeric substrates. To investigate this, we have been assessing the effects of PIF1 and RecQ helicases on telomerase activity *in vitro*, starting with Hrq1 and Pif1 [28,29]. Here, we extended this line of inquiry to include recombinant Sgs1. As shown in Fig. 2A, Sgs1 significantly stimulated *in vitro* telomerase activity at all helicase concentrations tested ($p < 0.001$). This is in contrast to Pif1, which stimulates telomerase at low concentrations and becomes a potent inhibitor at high concentrations, as well as Hrq1, which stimulates telomerase only at high concentrations (Fig. 2 and [28]). Although Sgs1 and its human homolog BLM are known to support telomere maintenance by regulating recombination at telomeres [30,31,62–64,66–70], there are no reports of a direct interaction between either of these helicases and the telomerase holoenzyme or direct regulation of telomerase activity. As we reported previously though, measuring the effects of a single helicase on telomerase *in vitro* does not fully recapitulate the *in vivo* milieu where multiple helicases localize to and act on telomeric DNA [28].

Therefore, we next tested the effects of combinations of helicases on *in vitro* telomerase activity. Fig. 2B is a representative image of one such assay, and Fig. 2C compares the total telomerase activity quantified in these assays across three experiments. As shown previously, the combination of Hrq1 and Pif1 is a strong telomerase inhibitor, regardless of the catalytic activity of Hrq1, but combining inactive Pif1 (Pif1-K264A) with active Hrq1 yields a strong stimulation of telomerase (Fig. 2C). When both Hrq1 and Pif1 are inactive, there is no effect on telomerase *in vitro*. In reactions containing Sgs1 and either Hrq1 or Hrq1-K318A, telomerase is stimulated to levels similar to that of adding Sgs1 alone (Fig. 2). In contrast, the combination of Pif1 and Sgs1 inhibits telomerase activity to Pif1-alone levels. However, Pif1 catalytic activity was necessary for the inhibition, as the combination of Sgs1 and Pif1-K264A stimulated telomerase activity like Sgs1 alone.

It should be noted that in reactions containing single helicases, the helicase concentration is 360 nM. In reactions containing two helicases, the total helicase concentration was held at 360 nM, meaning that each helicase was added at only 180 nM. Therefore, even in cases where it appears that the addition of a second helicase has no effect on *in vitro* telomerase activity relative to a single-helicase reaction (e.g., Pif1 + Sgs1 vs. Pif1; Fig. 2C) the measured telomerase activity may actually be different when considering the effect of 180 nM of either of the helicase alone. Thus, future work should investigate how *in vitro* telomerase activity is affected by combinations of Hrq1, Pif1, and Sgs1 across a broad range of helicase concentrations. Further, determining the effects of the *sgs1-K706A* allele on how GCR events are healed, how telomere length in the presence and absence of active Hrq1 and Pif1 changes, and how *in vitro* telomerase activity alone and in combination with Hrq1 and Pif1 is affected will be needed to address how wild-type Sgs1 may be regulating telomerase activity.

4. Conclusions

The methodology described here can be used to assess the effects of not just DNA helicase mutants on genome integrity but any protein known or suspected to be involved in genome maintenance. When contemplating which alleles to investigate, one should consider not just full gene deletions but mutations that lead to truncations, frameshifts, and/or inactivation of proteins. This is especially true of *S. cerevisiae* genes with disease-linked human homologs because many clinical alleles are not clean deletions of genes but, rather, yield crippled protein (e.g., [71]). Additionally, alterations in protein levels, including both up- and downregulation in some cases [72], can result in a disease state, so investigating genome integrity through the lens of transcription levels and protein stability is further warranted. As post-translational modifications are known to affect DNA helicase activities (e.g., [72,73]), they should also be considered. Finally, at least in the case of GCR assays,

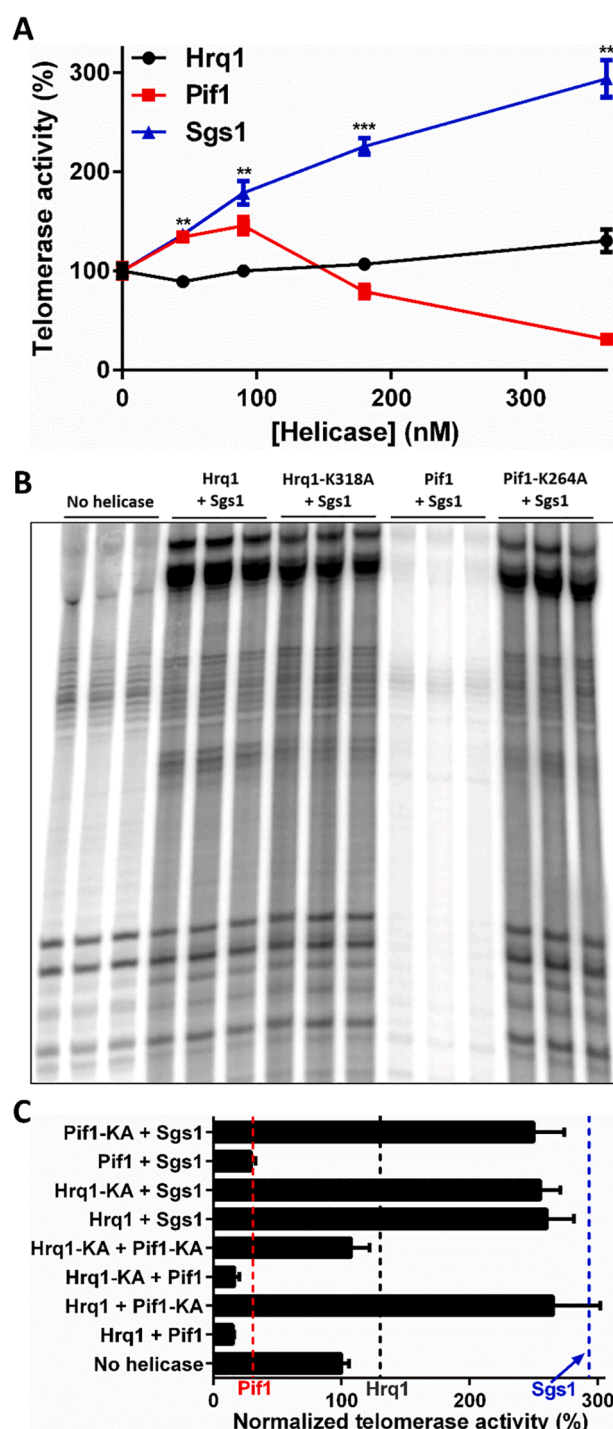


Fig. 2. Sgs1 stimulates telomerase activity *in vitro*. **A)** The effects of increasing concentrations of Hrql, Pif1, or Sgs1 on *in vitro* telomerase activity are shown. Significant differences were determined in GraphPad Prism by multiple t tests using the Holm-Sidak method, with $\alpha = 5\%$ and without assuming a consistent S.D. ($n = 3$). **, $p < 0.001$; and ***, $p < 0.0001$ relative to reactions lacking added helicase. **B)** Representative gel image of telomerase activity on the Tel50 substrate [28] in the absence and presence of 360 nM added helicases. Each set of three lanes shows independent experimental repeats of the same reaction conditions. **C)** Quantification of the overall telomerase activity (total signal in each lane, minus the loading control; not shown) measured in the absence and presence of the indicated helicase pairs. All values were normalized to the “No helicase” control = 100%. The dashed lines represent telomerase activity levels in the presence of the indicated helicases alone.

experiments can be extended to the chemical genetics realm by exposing cultures to small molecule drugs to assess their effects on genome stability in various mutant backgrounds (e.g., [9]).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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